1.25 Graph Sketching: e and Natural Logarithms

Worked Solutions

1 a)

$y = e^x$.

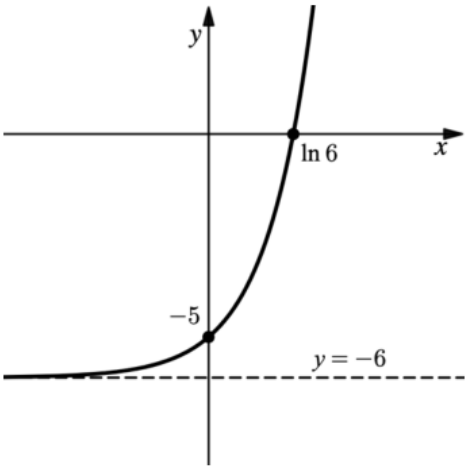
1 b)

$y = e^x + 5$.

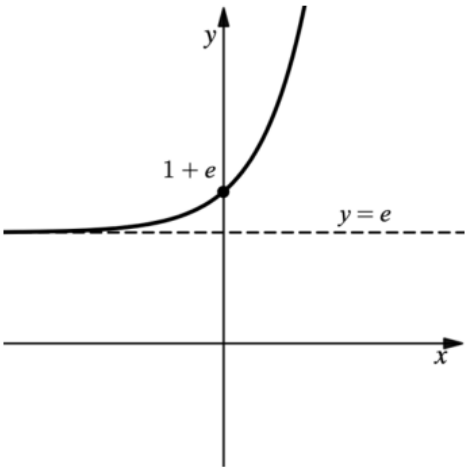
1 c)

$y = e^x - 4$.

2 a)



2 b)



3 a)

$y = e^{x-a} + 3$
$(0, e^{-1} + 3) : e^{-1} + 3 = e^{-a} + 3 \Rightarrow a = 1$
$y = e^{x-1} + 3.$

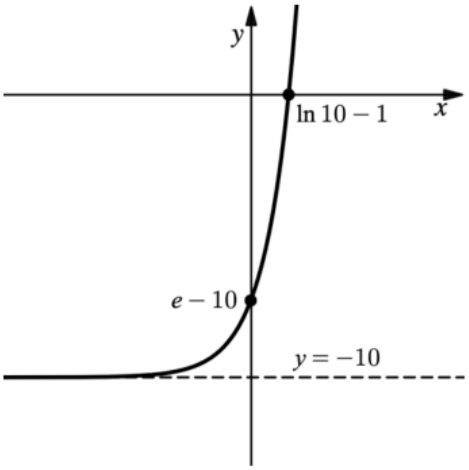
3 b)

$y = e^{x-a} - 5$
$(0, e^{2} - 5) : e^{2} - 5 = e^{-a} - 5 \Rightarrow a = -2$
$y = e^{x+2} - 5.$

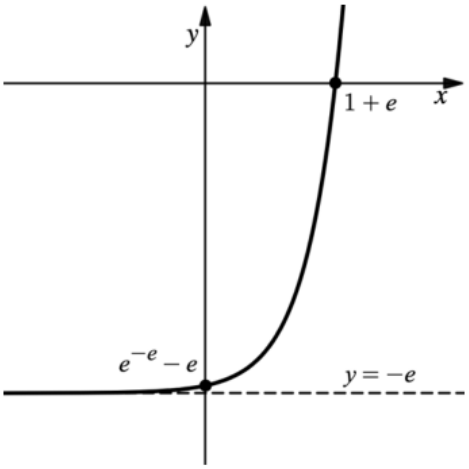
3 c)

$y = e^{x-a} - 1$
$(0, e^{-3} - 1) : e^{-3} - 1 = e^{-a} - 1 \Rightarrow a = 3$
$y = e^{x-3} - 1.$

4 a)



4 b)



5 a)

$y = e^{a-x} + 2$
$(0, e + 2) : e + 2 = e^a + 2 \Rightarrow a = 1$
$y = e^{1-x} + 2.$

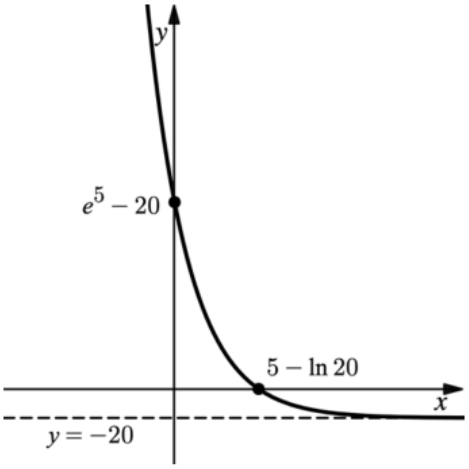
5 b)

$y = e^{a-x} - 3$
$(0, e^2 - 3) : e^2 - 3 = e^a - 3 \Rightarrow a = 2$
$y = e^{2-x} - 3.$

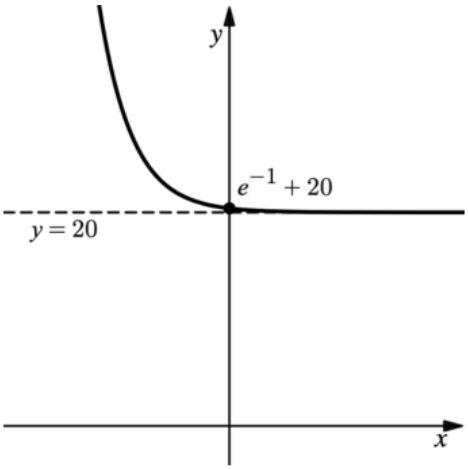
5 c)

$y = e^{a-x} - 9$
$(0, e^{-3} - 9) : e^{-3} - 9 = e^a - 9 \Rightarrow a = -3$
$y = e^{-3-x} - 9.$

6 a)



6 b)



7 a)

$y = 1 - e^{x-a}$
$(0,0) : 0 = 1 - e^{-a} \Rightarrow e^{-a} = 1 \Rightarrow a = 0$
$y = 1 - e^{x}$.

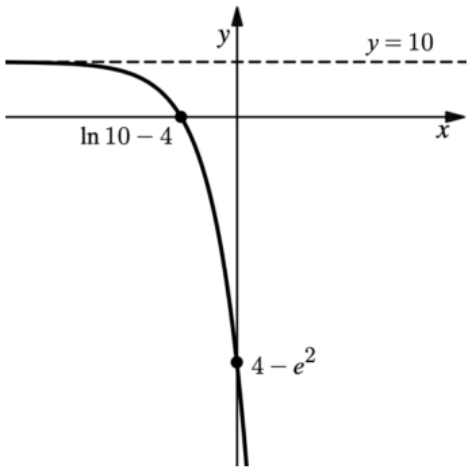
7 b)

$y = -2 - e^{x-a}$
$(0, -2 - e^{-1}) : -2 - e^{-1} = -2 - e^{-a} \Rightarrow a = 1$
$y = -2 - e^{x-1}$.

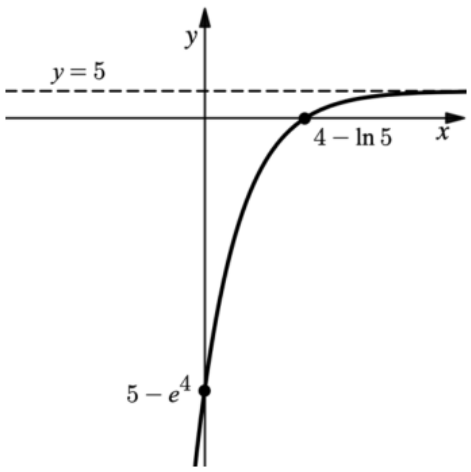
7 c)

$y = 4 - e^{x-a}$
$(0, 4 - e^{2}) : 4 - e^{2} = 4 - e^{-a} \Rightarrow a = -2$
$y = 4 - e^{x+2}$.

8 a)



8 b)



9 a)

$y = \ln(x - 3)$.

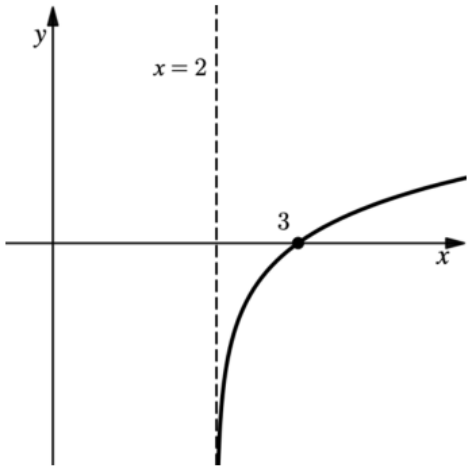
9 b)

$y = \ln(x + 5)$.

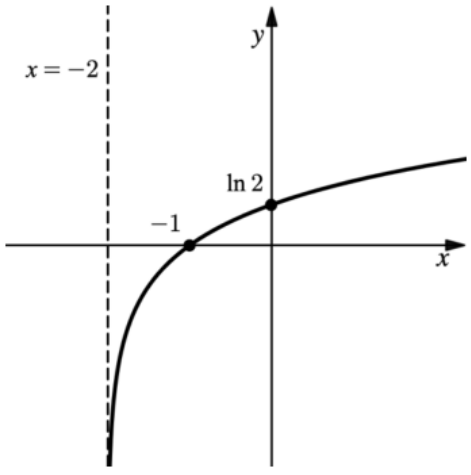
9 c)

$y = \ln(x + 1)$.

10 a)



10 b)



11 a)

$y = \ln x + b$
$(e^{-1}, 0) : 0 = \ln e^{-1} + b \Rightarrow 0 = -1 + b \Rightarrow b = 1$
$y = \ln x + 1$.

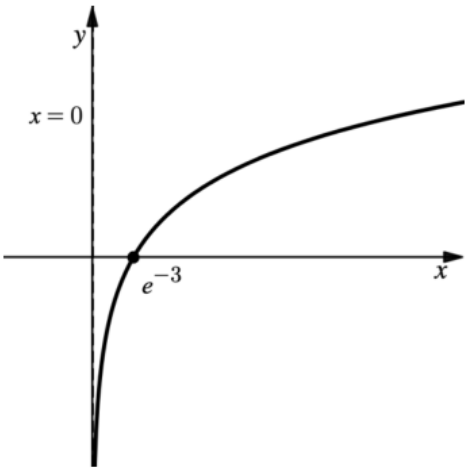
11 b)

$y = \ln x + b$
$(e^2, 0) : 0 = \ln e^2 + b \Rightarrow 0 = 2 + b \Rightarrow b = -2$
$y = \ln x - 2.$

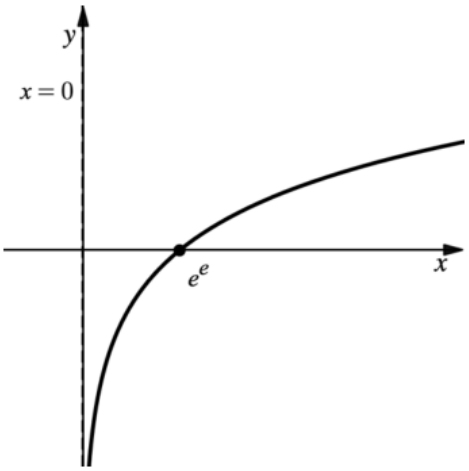
11 c)

$y = \ln x + b$
$(e^{-\frac{1}{2}}, 0) : 0 = \ln e^{-\frac{1}{2}} + b \Rightarrow 0 = -\frac{1}{2} + b \Rightarrow b = \frac{1}{2}$
$y = \ln x + \frac{1}{2}.$

12 a)



12 b)



13 a)

$y = \ln(x - 1) + b$
$(e^{-1} + 1, 0) : 0 = \ln(e^{-1} + 1 - 1) + b$
$\Rightarrow 0 = \ln(e^{-1}) + b \Rightarrow 0 = -1 + b \Rightarrow b = 1$
$y = \ln(x - 1) + 1.$

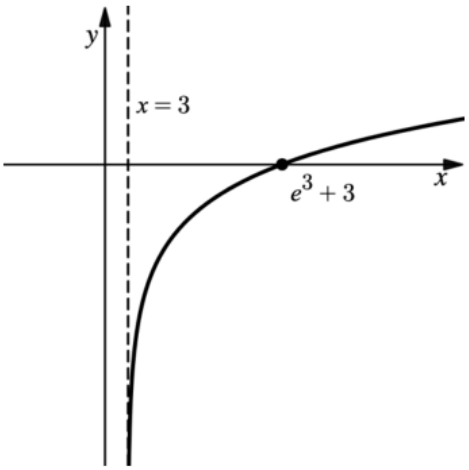
13 b)

$y = \ln(x+2) + b$
$(e^{-3} - 2, 0) : 0 = \ln(e^{-3} - 2 + 2) + b$
$\Rightarrow 0 = \ln(e^{-3}) + b \Rightarrow 0 = -3 + b \Rightarrow b = 3$
$y = \ln(x+2) + 3.$

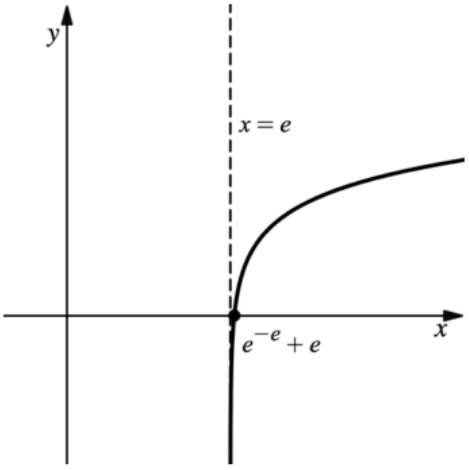
13 c)

$y = \ln(x+1) + b$
$(e^2 - 1, 0) : 0 = \ln(e^2 - 1 + 1) + b$
$\Rightarrow 0 = \ln(e^2) + b \Rightarrow 0 = 2 + b \Rightarrow b = -2$
$y = \ln(x+1) - 2.$

14 a)



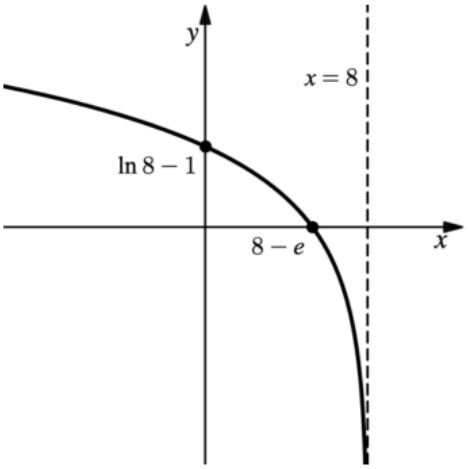
14 b)



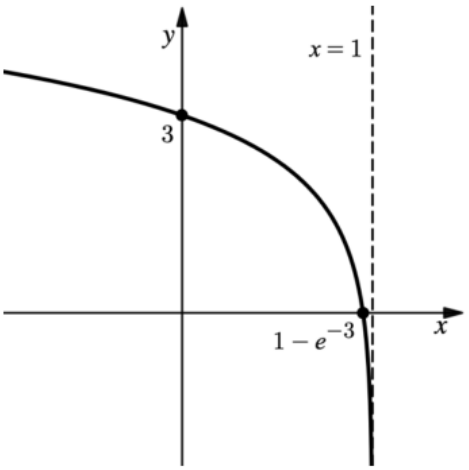
15)

$y = \ln(5-x) + b$
$(5 - e^{-2}, 0) : 0 = \ln(5 - (5 - e^{-2})) + b$
$\Rightarrow 0 = \ln(e^{-2}) + b \Rightarrow 0 = -2 + b \Rightarrow b = 2$
$y = \ln(5-x) + 2.$

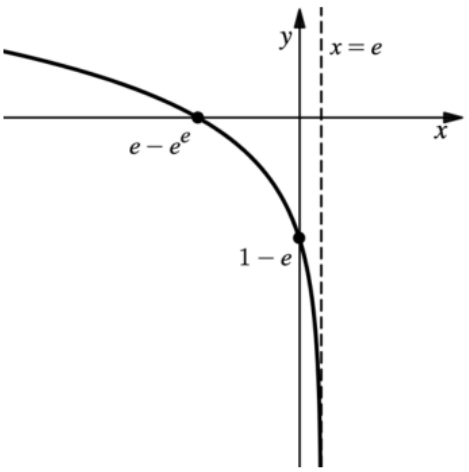
16 a)



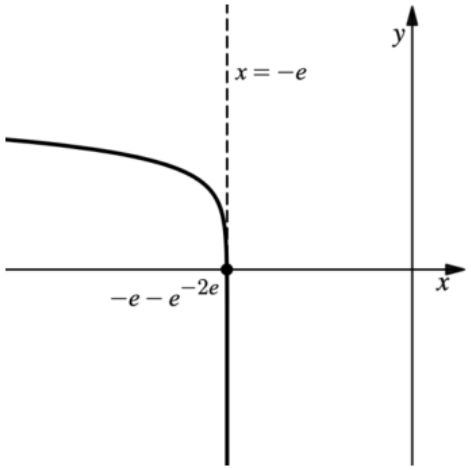
16 b)



16 c)



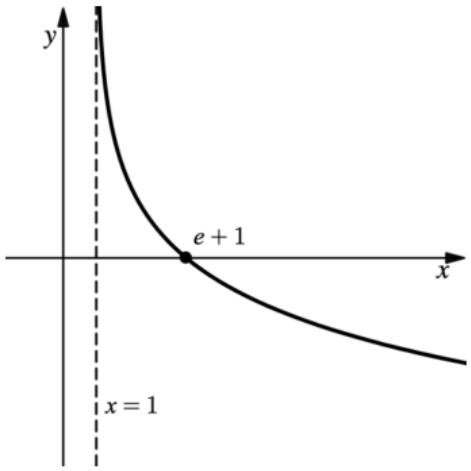
16 d)



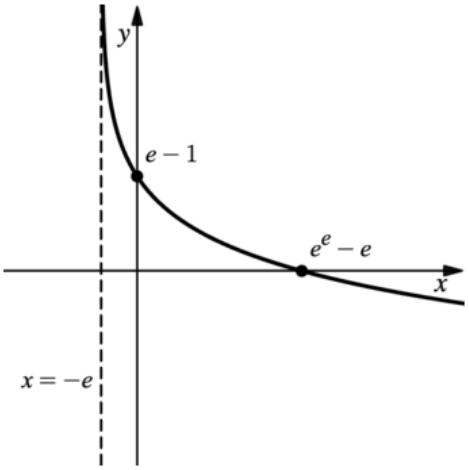
17)

$y = b - \ln(x-2)$
$(e^3+2, 0) : 0 = b - \ln(e^3+2-2)$
$\Rightarrow 0 = b - \ln(e^3) \Rightarrow 0 = b - 3 \Rightarrow b = 3$
$y = 3 - \ln(x-2)$.

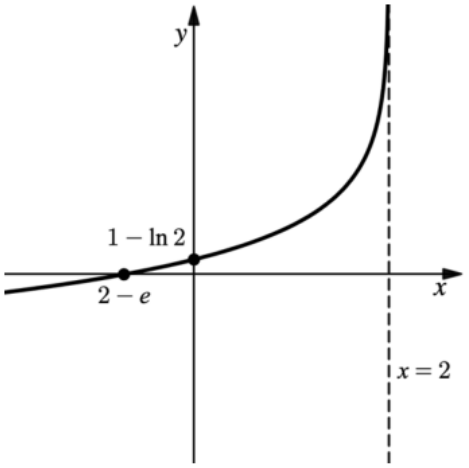
18 a)



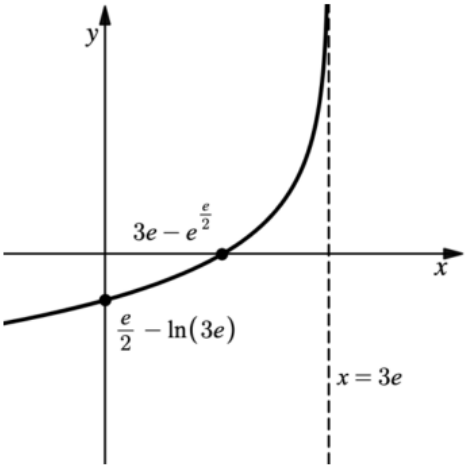
18 b)



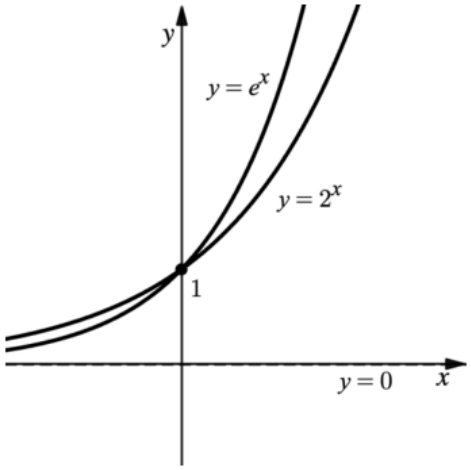
18 c)



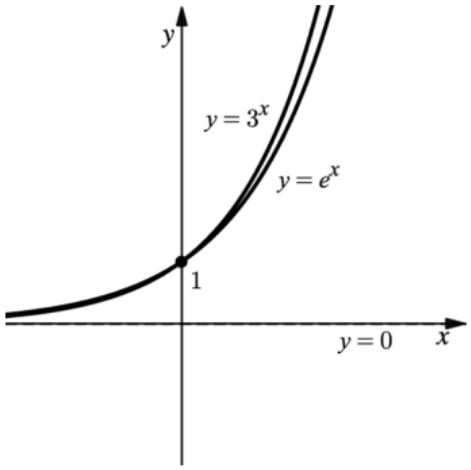
18 d)



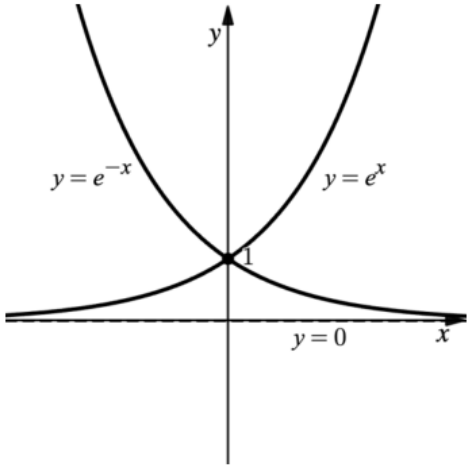
19 a)



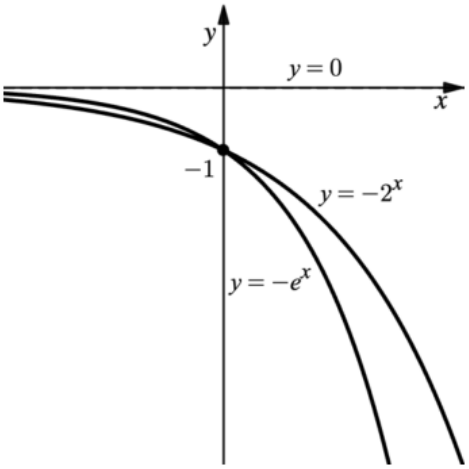
19 b)



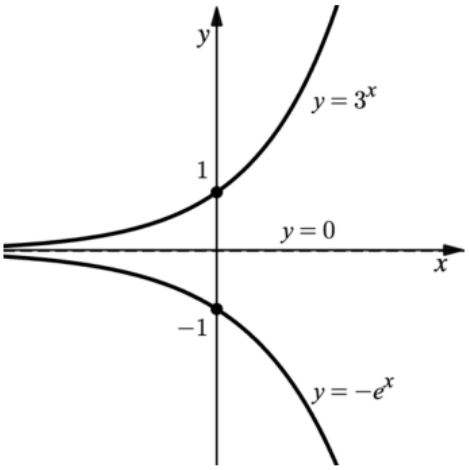
19 c)



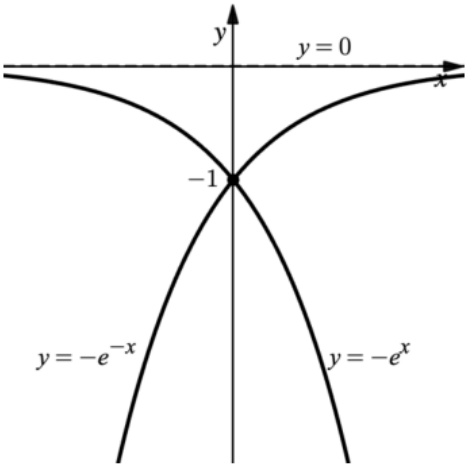
20 a)



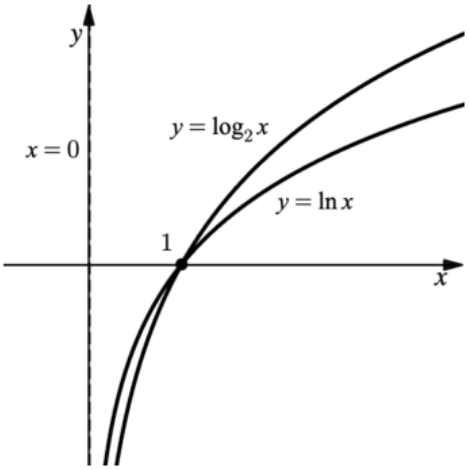
20 b)



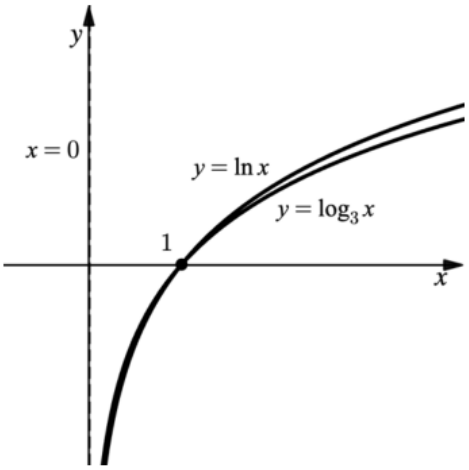
20 c)



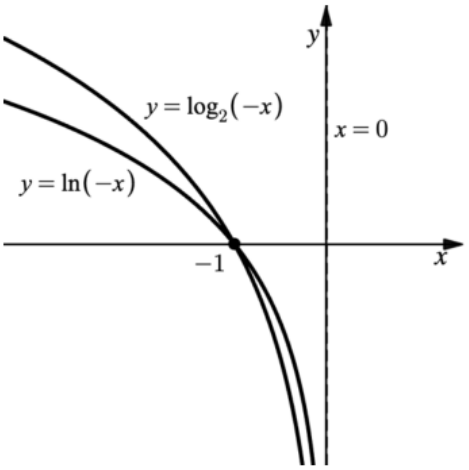
21 a)



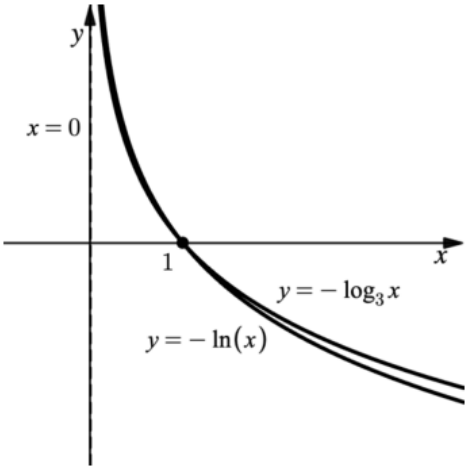
21 b)



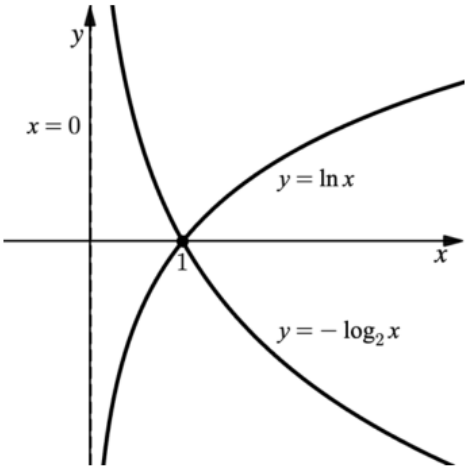
21 c)



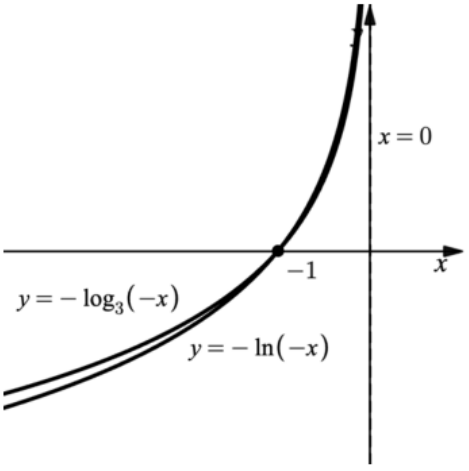
22 a)



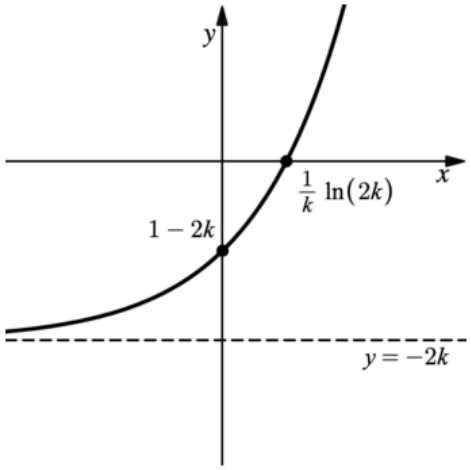
22 b)



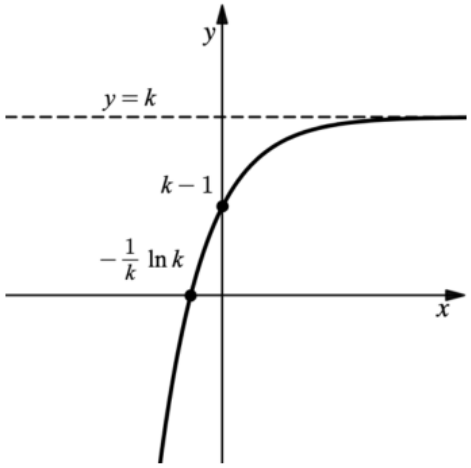
22 c)



23 a)



23 b)



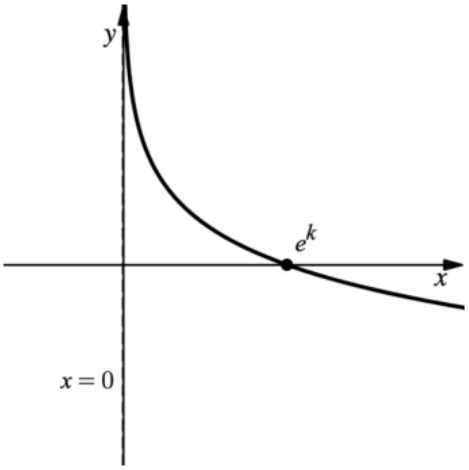
24 a)

$y = e^{-\frac{x}{k}} - k.$

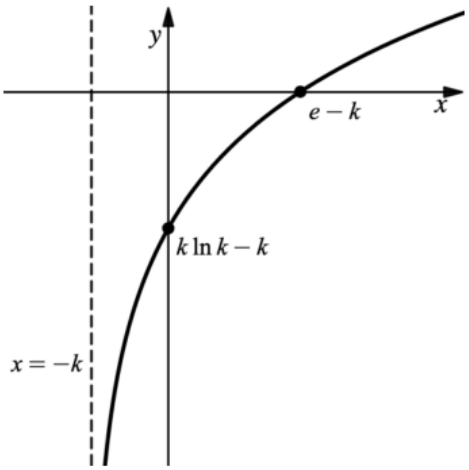
24 b)

$y = 2k - e^{\frac{x}{2k}}.$

25 a)



25 b)



26 a)

$y = k - \ln(x + k)$.

26 b)

$y = \ln(k - x) - \ln k$.